An Overview of the Core Deterministic ODE Model

The ODE model represents the rate of change of these six quantities:

- $SMO_{cytoplasm}$ - the concentration fo Smoothened(SMO) in the cytoplasm of the cell
- $SMO_{cillium}$ - the concentration of ciliary SMO
- $SMO_{ubiquitinated}$ - the concentration of ubiquitinated SMO
- MMM - the concentration of the MMM complex(MGRN1, MEGF8, MOSMO)
- $MMM_{non-functional}$ - the concentration of non-functional MMM complex after it binds to the de-novo protein.

The concentrations of the De-Novo($[De-Novo]$) protein and the linker protein($Linker$) are assumed to be constant. The equations describing the rate of change of the six aforementioned quantities are shown below:

$$\begin{aligned}\frac{d(SMO_{cytoplasm})}{dt} &= [k_{form,SMO_{cytoplasm}} + k_{out}[SMO_{cilium}]] \\ &\quad -(k_{deg,SMO_{cytoplasm}} + k_{in}[SMO_{cytoplasm})\end{aligned} \tag{1}$$

$$\frac{dSMO_{cilium}}{dt} = k_{in}[SMO_{cilium}] - (k_{deg,SMO_{cilium}} + \frac{V_{max}[SMO_{cilium}]}{K_m + [SMO_{cilium}]}) \tag{2}$$

$$\frac{dSMO_{ubiquitinated}}{dt} = \frac{V_{max}[SMO_{cilium}]}{K_m + [SMO_{cilium}]} \tag{3}$$

$$\begin{aligned}\frac{d(MMM)}{dt} &= (k_3[MMM_{non-functional}] + k_{form,MMM} + k_{off}[MMM - Linker]) \\ &\quad - (k_3[MMM][DeNovo] + k_{deg,MMM} + k_{form,MMM-Linker}[MMM][Linker])\end{aligned} \tag{4}$$

$$\frac{d(MMM_{non-functional})}{dt} = k_3[MMM][De-Novo] - k_{-3}[MMM_{non-functional}] \tag{5}$$

In equation 2, $V_{max} = k_{cat}[MMM]$, where k_{cat} is the rate constant for the formation of SMOubiquitinated (Product) from MMM-$SMO_{cillium}$(Enzyme-Substrate) complex. K_m is defined as the Michelis Menten constant, and is defined by the equation shown below:

$$K_m = \frac{k_{-1} + k_{cat}}{k_1}$$

Where:

- k_1 is the rate of reaction of the formation of the MMM-$SMO_{cillium}$
- k_{-1} is the rate of the reverse reaction where MMM-$SMO_{cillium}$ complex(Enzyme-Substrate complex) breaks down into MMM(Enzyme) and $SMO_{cillium}$ (Substrate)

I introduced a new equation to model the rate of the MMM-Linker complex, shown below:

$$\frac{d(MMM - LinkerComplex)}{dt} = k_{f,MMM}[MMM][Linker] - k_{off}[MMM - LinkerComplex] \tag{6}$$

This equation assumes that the MMM-Linker complex follows the quasi-steady-state assumption. The change in the MMM-Linker complex only depends on the concentrations of Linkers and MMM, the rate of system-mediated formation, and the rate of system-mediated degradation.

The paper *Affinity and cooperativity modulate ternary complex formation to drive targeted protein degradation* published in *Nature* models the rate of reaction PROTAC-like systems using the equation below:

$$\frac{v}{V_{max}} = \frac{\alpha}{\alpha + (\frac{2(K_{LP}+[P]_t}{[P]_t+[T]_t+K_{T_p}-\sqrt{([P]_t+[T]_t+K_{T_p})^2-4[P_t][T_t]}})}$$

Where:

- v is the initial rate of target ubiquitination
- V_{max} is the maximum velocity, and the product of the tertiary complex breakdown rate constant and total ligase concentration
- $[P]_t$ is the PROTAC concentration at time t, which in this case is the Linker concentration
- $[T]_t$ is the target molecule concentration at time t, which in our case is the concentration of SMO
- K_{TP} represents the rate of PROTAC and target binding- in this case the rate of Linker-SMO binding.
- K_{LP} is rate constant of ligase and PROTAC binding - in this case the rate of MMM-Linker binding
- α is the cooperativity coefficient of the reaction. This number represents how the binding of one ligand would influence the affinity of further binding to occur.

If this paper is used as a reference, then Equation 3 describing the rate of change of Ubiquitinated SMO concentrations over time can be re-written as shown below.

$$\frac{d(SMO_{ubiquitinated}}{dt} = \frac{V_{max}\alpha}{\alpha + (\frac{2(K_{MMM-Linker}+[Linker]_t}{[Linker]_t+[SMO]_t+K_{SMO-Linker}-\sqrt{([Linker]_t+[SMO]_t+K_{SMO-Linker})^2-4[Linker]_t[SMO]_t}})}$$

Based on this revision, Equation 2 describing the rate of change of ciliary SMO over time can be rewritten as shown below.

$$\frac{dSMO_{cilium}}{dt} = k_{in}[SMO_{cilium}] - (k_{deg,SMO_{cilium}} + \frac{V_{max}\alpha}{\alpha + (\frac{2(K_{MMM-Linker}+[Linker]_t}{[Linker]_t+[SMO]_t+K_{SMO-Linker}-\sqrt{([Linker]_t+[SMO]_t+K_{SMO-Linker})^2-4[Linker]_t[SMO]_t}})})$$